Data Management for Machine Learning Assignment-2

Title : *End-to-End Data Management Pipeline for a Recommendation System*
Dataset : *MovieLens 1M*
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GitHub Link: https://github.com/mudassar-hamid/recommendation-pipeline

1. Objective: Design and implement an end-to-end data management pipeline for a recommendation system.

2. Overview: This project is a modular, reproducible pipeline that takes raw user, item and interaction data to a tracked and evaluated recommender: The project implements a reproducible movie recommendation pipeline covering data ingestion, automated validation, chronological preparation, feature generation, two recommendation models, ranking evaluation, experiment tracking and Prefect orchestration.

3. Dataset: The source dataset is MovieLens 1M from GroupLens. It contains user demographic fields, movie titles/genres and timestamped ratings. Age is retained as the dataset's age-group code, not interpreted as an exact age.

- Age is chosen from the following ranges:

1: "Under 18"

18: "18-24"

25: "25-34"

35: "35-44"

45: "45-49"

50: "50-55"

56: "56+"

The dataset has three different data files.

1. **users.dat**
   Fields:

UserID	Gender	Age	Occupation	Zip-code

2. **movies.dat**
   Fields:

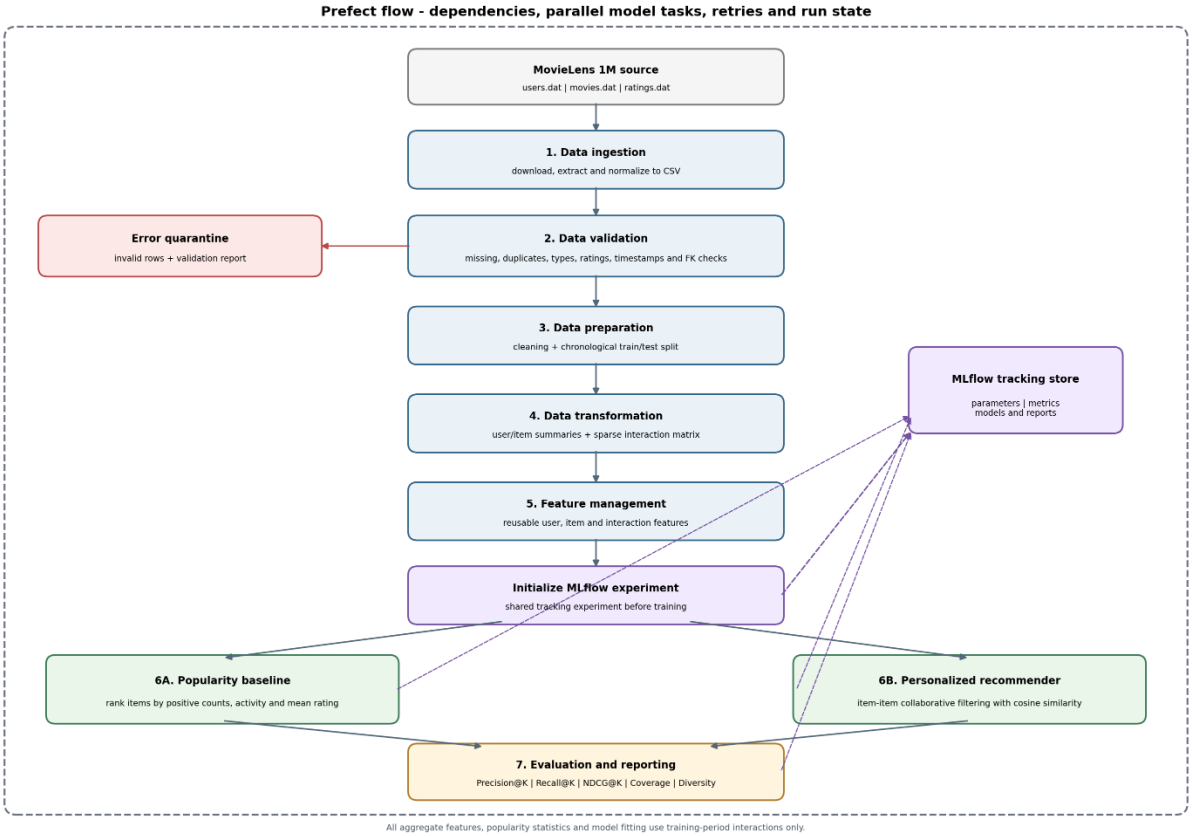
MovieID	Title	Genres

3. **ratings.dat**
   Fields:

UserID	MovieID	Rating	Timestamp

4. Pipeline design: The workflow downloads and normalizes the original MovieLens DAT files into three CSV entities: users, items and interactions. Validation checks missing values, duplicate entity/event keys, numeric types, rating range (1–5), timestamp validity and user/item referential integrity. Invalid rows are quarantined under data/errors and valid records continue. Preparation uses a global chronological 80/20 split; all derived summaries, features and models use training interactions only.

5. Architecture:



6. Design decisions and tools:

Ingestion	MovieLens 1M from GroupLens read as text (types are checked later). The source is already clean. It contains user demographic fields, movie titles/genres and timestamped ratings.
Validation	Pandera schemas (nulls, ranges, allowed values) plus relational checks (foreign keys, duplicates). Invalid rows go to an error dataset with the failed checks instead of being dropped silently; a quality gate stops the run above 5% errors and is not retried.
Preparation	Standardized text and dtypes, UTC event time, one release year imputed (and flagged), empty columns dropped.
Split & leakage	One global time-based cutoff (2000-12-02T14:52:18+00:00): train before, test after, so no future data reaches the training set. Derived datasets and all features use training data only; the split is checked for temporal leakage.

Features	16 documented features for users, items and user-item pairs. Training, evaluation and prediction read the same stored features; a contract check fails the run if materialized columns differ from the definitions. Label-derived features are marked training-only. A feature catalog 'feature_catalog' is saved in CSV and Markdown.
Models	(1) popularity baseline (2) Personalized recommender
Tracking	MLflow (SQLite): parameters, metrics and artifacts per model run, plus a CSV export
Orchestration	Prefect flow: dependencies are task arguments, both models train in parallel (fan-out/fan-in), transient failures are Retried (screenshot of a successful run: reports/orchestration_run_screenshot.png). A plain sequential runner shares the same stage functions.
Reproducibility	Prefect orchestrates ingestion ® validation ® preparation ® transformation ® feature generation ® model training ® evaluation/reporting. Task retries are configured, and the two models are trained as separate tasks. MLflow records experiment parameters, metrics and artifacts using a local tracking store. The full workflow is launched with one command, documented in README.md.

Data quality and preparation:

Dataset	Checked	Valid	Quarantined
Users	6040	6040	0
Items	3883	3883	0
Interactions	1000214	1000207	7

The chronological cutoff is 2000-12-02T14:52:18+00:00. Training contains 800,168 rows (80.0%); test contains 200,041 rows.

Feature management: A reusable feature layer exports user features (activity count, average rating, preferred genre, recency and source demographics), item features (genre, interaction popularity, positive interaction count and average rating), and training interaction features (rating, positive-feedback flag, UTC weekday/hour and recency relative to the training cutoff). A feature catalog is saved in CSV and Markdown.

feature_name	entity	description	data_type	computation_or_source
user_activity_count	user	Number of training interactions for the user	integer	Count of rows in train_interactions
user_avg_rating	user	Mean rating supplied by the user in training data	float	Mean rating grouped by user
preferred_category	user	Most frequently interacted movie genre in training data	string	Most common genre among movies interacted with by the user
days_since_last_interaction	user	Days from the training cutoff to the user's latest training interaction	float	Training cutoff minus user last interaction
gender	user	MovieLens user gender field	category	Source demographic field
age_group_code	user	MovieLens age bucket code; not an exact age	integer	Source demographic field
occupation_code	user	MovieLens occupation category code	integer	Source demographic field
item_interaction_count	item	Number of training ratings for the item	integer	Count of training interactions grouped by item
item_positive_interaction_count	item	Number of training ratings at or above the positive threshold	integer	Count of ratings >= configured threshold
item_avg_rating	item	Mean training rating for the item	float	Mean rating grouped by item
genres	item	Pipe-separated movie genres	string	MovieLens item metadata
interaction_rating	interaction	Observed training rating	float	Source interaction value
interaction_is_positive	interaction	Whether rating meets the positive feedback threshold	boolean	rating >= configured threshold
interaction_weekday	interaction	UTC weekday of interaction	integer	Timestamp weekday (Monday=0)
interaction_hour_utc	interaction	UTC hour of interaction	integer	Timestamp hour in UTC
days_before_training_cutoff	interaction	Days between event timestamp and training cutoff	float	Training cutoff minus event timestamp

Model training, evaluation and orchestration

Recommendation models: The project implements two recommendation models.

Model 1: Popularity-based recommender (baseline)

This is the baseline model.

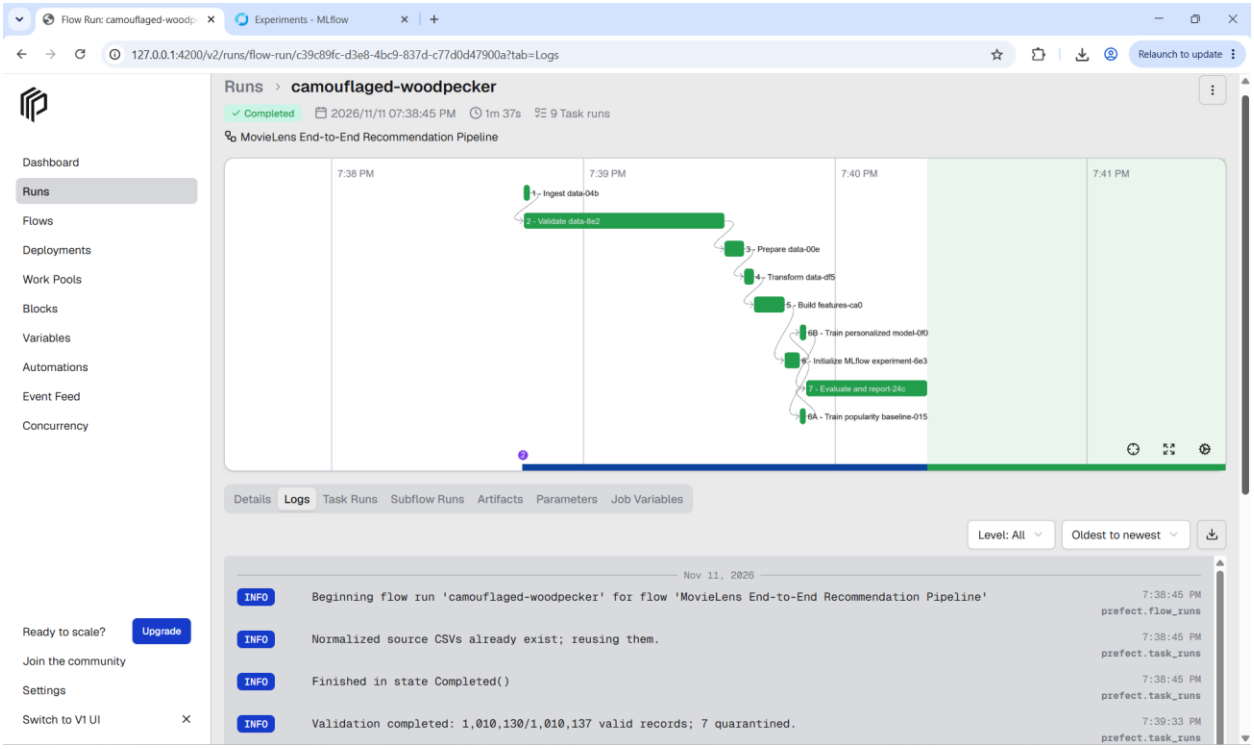
It recommends the most popular items to every user, regardless of individual preferences. The baseline ranks items by training positive-rating count, then total training interactions and average rating.

Model 2: Personalized recommender

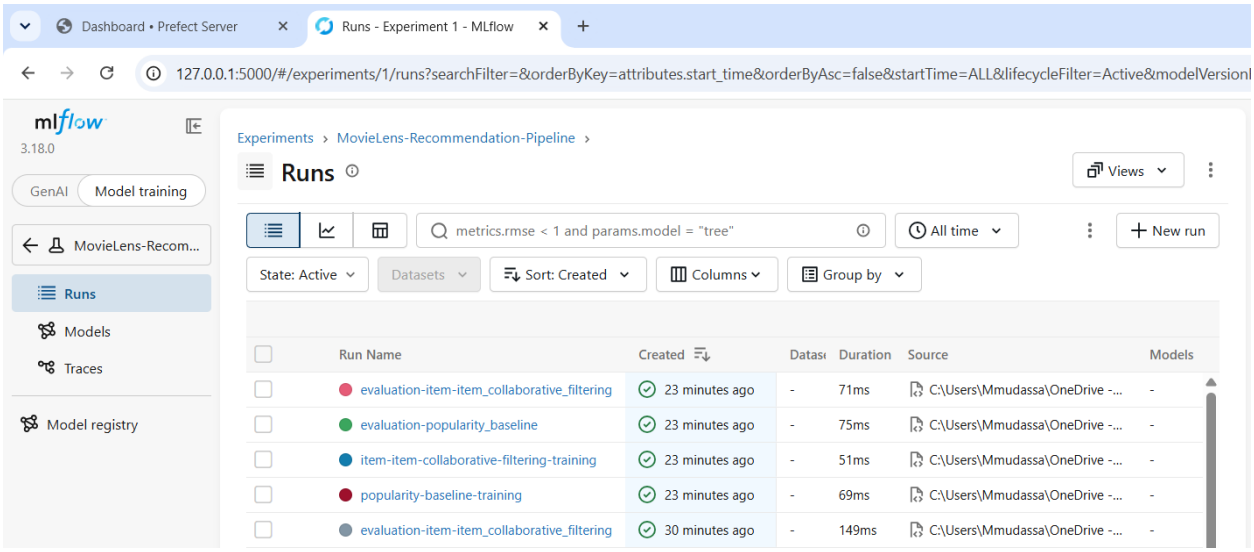
The second model must personalize recommendations using historical user interactions or item information. The personalized model is item-item collaborative filtering: cosine similarity is computed between item vectors from a binary matrix of positive training ratings. Previously seen items are removed from each user's candidate list; popularity is used as a fallback/tie-breaker.

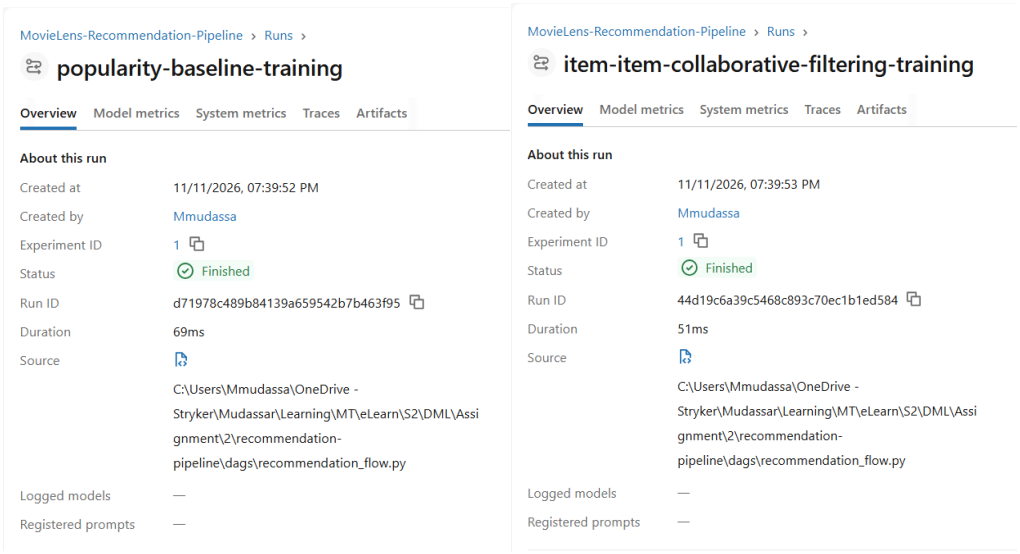
Execution:

```
(global_gpu_env) PS C:\Users\Mmudassa\OneDrive - Stryker\Mudassar\Learning\MT\eLearn\S2\DML\Assignment\2\recommendation-pipeline> python -m dags.recommendation_flow
19:38:45.612 | INFO    | Flow run 'camouflaged-woodpecker' - Beginning flow run 'camouflaged-woodpecker' for flow 'MovieLens End-to-End Recommendation Pipeline'
19:38:45.626 | INFO    | Flow run 'camouflaged-woodpecker' - View at http://127.0.0.1:4200/runs/flow-run/c39c89fc-d3e8-4bc9-837d-c77d0d47900a
19:38:45.898 | INFO    | Task run '1 - Ingest data-04b' - Normalized source CSVs already exist; reusing them.
19:38:45.902 | INFO    | Task run '1 - Ingest data-04b' - Finished in state Completed()
19:39:33.560 | INFO    | Task run '2 - Validate data-8e2' - Validation completed: 1,010,130/1,010,137 valid records; 7 quarantined.
19:39:33.704 | INFO    | Task run '2 - Validate data-8e2' - Finished in state Completed()
19:39:38.369 | INFO    | Task run '3 - Prepare data-00e' - Prepared 800,168 train and 200,039 test interactions; cutoff=2000-12-02 14:52:18+00:00.
19:39:38.379 | INFO    | Task run '3 - Prepare data-00e' - Finished in state Completed()
19:39:40.718 | INFO    | Task run '4 - Transform data-df5' - Created user activity summary, item popularity and matrix (6040, 3883) (800,168 ratings).
19:39:40.728 | INFO    | Task run '4 - Transform data-df5' - Finished in state Completed()
19:39:47.933 | INFO    | Task run '5 - Build features-ca0' - Built feature tables: users=5,400, items=3,883, interactions=800,168.
19:39:48.023 | INFO    | Task run '5 - Build features-ca0' - Finished in state Completed()
19:39:50.323 | INFO    | Task run '6 - Initialize MLflow experiment-6e3' - MLflow Tracking URI: sqlite:///mlflow.db
19:39:50.365 | INFO    | Task run '6 - Initialize MLflow experiment-6e3' - Actual MLflow URI: sqlite:///mlflow.db
19:39:51.656 | INFO    | Task run '6 - Initialize MLflow experiment-6e3' - MLflow experiment ready: MovieLens-Recommendation-Pipeline
19:39:51.662 | INFO    | Task run '6 - Initialize MLflow experiment-6e3' - Finished in state Completed()
19:39:52.808 | INFO    | Task run '6A - Train popularity baseline-015' - Trained popularity baseline for 3,662 items.
19:39:52.821 | INFO    | Task run '6A - Train popularity baseline-015' - Finished in state Completed()
19:39:53.151 | INFO    | Task run '6B - Train personalized model-0f0' - Trained item-item CF model: 3,662 items, 343,298 retained similarities.
19:39:53.166 | INFO    | Task run '6B - Train personalized model-0f0' - Finished in state Completed()
19:40:21.995 | INFO    | Task run '7 - Evaluate and report-24c' - Generated final report: C:\Users\Mmudassa\OneDrive - Stryker\Mudassar\Learning\MT\eLearn\S2\DML\Assignment\2\recommendation-pipeline\reports\final_report.pdf
19:40:21.997 | INFO    | Task run '7 - Evaluate and report-24c' - Model evaluation complete. Results:
19:40:22.001 | INFO    | Task run '7 - Evaluate and report-24c' -                          model  precision_at_5  recall_at_5  ndcg_at_5  coverage  diversity  users_evaluated  unique_items_recommended  catalog_size     rmse      mae  rating_predictions_evaluated
           Popularity baseline       0.231551     0.031395   0.241696  0.024577  0.746622          1122                      90         3662      NaN      NaN                             0
Item-item collaborative filtering       0.248307     0.038446   0.262408  0.123703  0.724950          1122                     453         3662 1.071359 0.82065                        104448
19:40:22.029 | INFO    | Task run '7 - Evaluate and report-24c' - Finished in state Completed()
19:40:22.032 | INFO    | Flow run 'camouflaged-woodpecker' - Recommendation pipeline completed successfully.
19:40:22.075 | INFO    | Flow run 'camouflaged-woodpecker' - Finished in state Completed()
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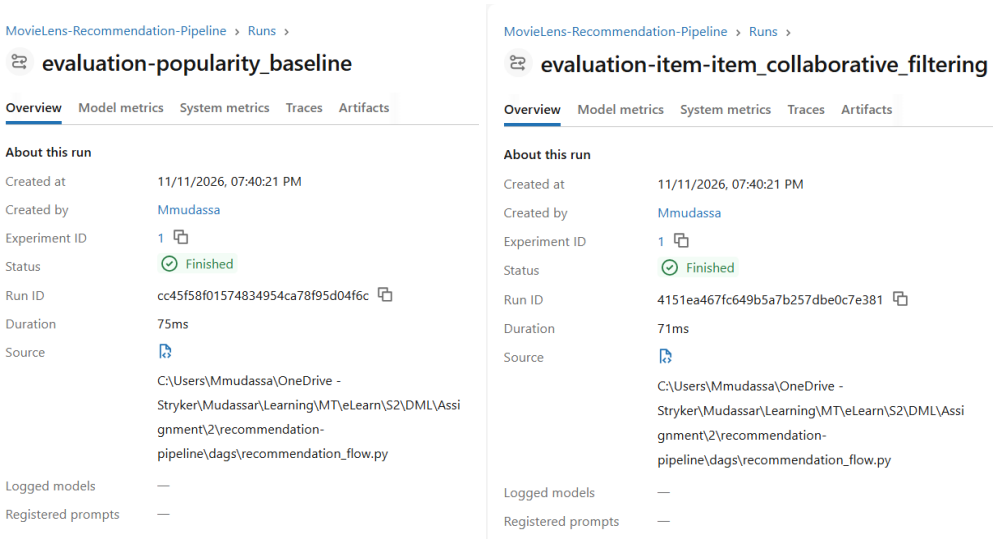


Prefect Run





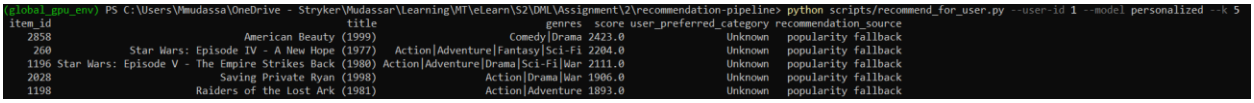
MLFlow Trainings



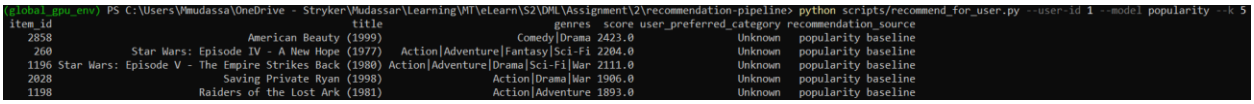
MLFlow Evaluation

Recommendations

1. Personalized recommendations:



2. Popularity-based recommendations:



Evaluation results:

Model	P@5	R@5	NDCG@5	Coverage	Diversity
Popularity baseline	0.2316	0.0314	0.2417	0.0246	0.7466
Item-item collaborative filtering	0.2483	0.0384	0.2624	0.1237	0.7250

Ranking metrics are macro-averaged across 1,122 users with at least one positive held-out rating for an item available in the training catalog. K=5; relevance is rating >= 4.0. Precision@K divides hits by K, Recall@K divides hits by the user's relevant held-out items, and NDCG@K discounts hits by rank. Coverage is the fraction of training catalog items recommended to at least one evaluated user. Diversity is mean intra-list Jaccard distance over genre sets.

Rating prediction evaluation (RMSE and MAE);

The personalized item-item model also estimates explicit 1–5 star ratings for held-out user–movie interactions. RMSE (root mean squared error) penalizes larger prediction errors more strongly, while MAE (mean absolute error) reports the average absolute difference between predicted and actual ratings. Lower values indicate more accurate rating predictions. These metrics are computed on held-out test ratings, not on ratings used to build the model.

Model	RMSE	MAE
Popularity baseline	N/A	N/A
Item-item collaborative filtering	1.0714	0.8207

N/A means that the model does not produce explicit user-specific rating predictions or that the corresponding metric was unavailable in the evaluation output.

Limitations and considerations

55 positive held-out interactions involving items absent from the training catalog are excluded from ranking evaluation, and 55,969 positive held-out interactions belonging to users without training history are separately counted. Neither model can recommend an unseen catalog item or personalize for a user with no history. The collaborative filter is item-item and uses genre metadata only for diversity measurement, not model scoring. New users/items therefore remain cold-start cases. The MovieLens 1M data reflects historical ratings and should not be interpreted as current viewing preferences. Results depend on the chosen chronological cutoff, positive-rating threshold and K.

Conclusion

The implementation demonstrates the complete data management lifecycle and compares a global popularity baseline with a personalized collaborative-filtering recommender. The saved validation logs, feature catalog, model artifacts, evaluation table, MLflow runs and Prefect execution history provide traceable evidence for the pipeline.